Total Printed Pages: 02

Roll No. :.....

MAB-101
Examination –Dec- 2025
B.Tech. I Sem : Common for all Branches
Linear Algebra and Integral Calculus

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

				BL	CO
Q.1	(a)	Write the statement of Taylor's theorem Theorem	02		
	(b)	Expand e^x cosx in power of x by Maclaurin's theorem.	02		
	(c)	. Find the radius of curvature of the following curves at the points indicated against them $\sqrt{x}+\sqrt{y}=\sqrt{a}$, (1/4, 1/4).	03		
	(d)	Discuss the maxima and minima of the function U = Sinx + Siny + Sin (x+y)	05		
		OR			
		Find the radius of curuature at the Pamt "t" of the curue . $x = a\cos^3 t \quad y = a\sin^3 t$	05		
2	(a)	If $u = \sin^{-1}(x/y) + \tan^{-1}(y/x)$ Show that $x\,\partial u/\partial x + y\,\partial u/\partial y = 0$	02		
	(b)	u=xy/x+y show that $x^2\frac{\partial^2 u}{\partial x^2}+y^2\frac{\partial^2 u}{\partial y^2}+2xy\frac{\partial^2 u}{\partial x\partial y}=0$	02		
	(c)	If $x^x y^y z^z = c$ then show that $\delta^2 z/\delta x\delta y = -(x \log ex)^{-1}$	03		
	(d)	(d) If u = f(r), Where $r^2 = x^2 + y^2$ Show that $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = f''(r) + 1/r\, f'(r)$.	05		
		OR			
		If $u=\sin^{-1}\left(\frac{x^{1/3}+y^{1/3}}{x^{1/2}+y^{1/2}}\right)^{1/2}$ then show that $x^2\frac{\partial^2 u}{\partial x^2}+y^2\frac{\partial^2 u}{\partial y^2}+2xy\frac{\partial^2 u}{\partial x\partial y}=\frac{\tan u}{144}(13+\tan^2 u)$	05		
	(a)	Ab initio Evaluate $\int_a^b e^x\,dx$	02		
	(b)	Find the limit as $n\to\infty$ of the series $\frac{1^2}{1^3+n^3}+\frac{2^2}{2^3+n^3}+\ldots+\frac{n^2}{n^3+n^3}$	02		

(c) Find the limit as $n \to \infty$ of the series

$$\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\left(1+\frac{3}{n}\right)\ldots\ldots\ldots\left(1+\frac{1}{n}\right)\right]^{1/n}$$

03

(d) **Evaluate** $\int_0^4 \int_0^x \int_0^{x+y} z\,dz\,dy\,dx$

05

OR

Change the order of integration in $\int_0^1 \int_y^1 x^2 \cos(x^2 - xy)\,dx\,dy$ and hence evaluate it.

05

Q.4 (a) Find the characteristic root's of the matrix $A=\begin{bmatrix}5 & 4\\ 1 & 2\end{bmatrix}$

02

(b) Define Echelon form of a matrix with example

02

(c) Reduce the Matrix to normal form and find it`s rank.

$$A = \begin{bmatrix}1 & -1 & 2 & -1\\ 4 & 2 & -1 & 2\\ 2 & 2 & -1 & 0\end{bmatrix}$$

$(5-\lambda)(2-\lambda)-4=0$

$\lambda^2-7\lambda+6=0$

$\lambda = \frac{7 \pm \sqrt{49-24}}{2} = \frac{7\pm 5}{2}$

$= 6, 1$

03

(d) Check the consistency of the following system of equations.

$$x_1+2x_2+3x_3 = 14$$
$$3x_1+x_2+2x_3 = 11$$
$$2x_1+3x_2+x_3 = 13$$

05

OR

Determine the eigen values and eigen vectors of the matrix

$$A=\begin{bmatrix}6 & -2 & 2\\ -2 & 3 & -1\\ 2 & -1 & 3\end{bmatrix}$$

05

Q.5 (a) Define Boolean algebra

02

(b) If p and q are two statements prove that $(p \wedge q) \Rightarrow q$ is a tautology

02

(c) State and prove Demorgan;s law

03

(d) Express the following function into disjunctive normal form:

$F(x, y, z) = x.y.z + (x+y)(x+z)$

05

OR

Simplify the following

05

(i) $(ab')(a'b)$

(ii) $abc + abc' + ab'c + a'bc$

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